

CO1 – QUESTION AND ANSWER

1. Write few typography CSS properties.

Some commonly used typography CSS properties are:

- font-family
- font-size
- font-weight
- font-style
- text-align
- line-height
- letter-spacing
- text-transform

2. Write few ways to apply color to text in a web page.

You can apply color to text using:

- **Named colors** → color: red;
- **Hex codes** → color: #ff0000;
- **RGB values** → color: rgb(255,0,0);
- **RGBA values** → color: rgba(255,0,0,0.7);
- **HSL values** → color: hsl(0,100%,50%);

3. List out the ways of applying CSS to an HTML page.

There are **three ways**:

1. **Inline CSS**
2. **Internal CSS (using <style> tag)**
3. **External CSS (using .css file linked with <link> tag)**

4. How will you link external CSS to an HTML page? Write that particular tag.

Use the <link> tag inside <head> section:

```
<link rel="stylesheet" href="style.css">
```

5. Explain about different selectors in CSS.

CSS selectors are patterns used to select HTML elements for applying styles.

1. **Tag (Element) Selector**

Selects all elements of a specific tag.

Example:

```
p { color: blue; }
```

2. **Class Selector**

Selects elements with a specific class name.

Example:

```
.box { padding: 10px; }
```

3. **ID Selector**

Selects an element with a unique ID.

Example:

```
#header { background: black; }
```

4. **Universal Selector**

Applies style to all elements.

```
* { margin: 0; }
```

5. **Group Selector**

Applies the same style to multiple elements.

```
h1, h2, h3 { font-family: Arial; }
```

6. **Descendant (Nested) Selector**

Applies style to elements inside other elements.

```
div p { color: green; }
```

7. **Attribute Selector**

Selects elements with specific attributes.

```
input[type="text"] { border: 1px solid black; }
```

6. What are the ways to apply CSS to an HTML page? Explain it.

There are **three ways** to apply CSS:

1. Inline CSS

- Written inside HTML tags using the style attribute.
- Used for quick, one-time styling.

```
<p style="color:red;">Hello</p>
```

2. Internal CSS

- Written inside <style> tag in the <head> section.
- Useful when a single page needs specific styling.

```
<head>
<style>
p { color: blue; }
</style>
</head>
```

3. External CSS

- Written in a separate .css file.
- Linked using <link> tag in HTML.
- Best method for large websites.

```
<link rel="stylesheet" href="style.css">
```

7. Write typography tag with internal, external and inline CSS and explain what will get higher priority and why?

Sample HTML with all three types of CSS

```
<!DOCTYPE html>
<html>
<head>
<style>
p { font-size: 20px; color: blue; }
</style>
<link rel="stylesheet" href="style.css">
</head>

<body>
<p style="color: red;">Typography Example</p>
</body>
</html>
```

style.css (external file)

```
p { color: green; }
```

Priority Order (Highest → Lowest)

1. **Inline CSS**

2. **Internal CSS**
3. **External CSS**
4. **Browser default styles**

Why Inline gets highest priority?

Because inline CSS is directly applied to the specific element, so it overrides internal and external styles unless !important is used.

8. Apply different CSS selectors (class, id, tag, nested, universal) in style.css for a sample HTML page you design.

Sample HTML Page

```
<!DOCTYPE html>

<html>

<head>
<link rel="stylesheet" href="style.css">
</head>

<body>

<h1 id="title">Welcome</h1>

<div class="container">
  <p>This is a paragraph inside container.</p>
</div>

<p class="info">This is another paragraph.</p>

</body>
</html>
```

style.css using all selectors

```
/* Universal selector */
* {
```

```
margin: 0;
padding: 0;
}
```

```
/* Tag selector */
p {
  font-size: 18px;
}
```

```
/* ID selector */
#title {
  color: blue;
  text-align: center;
}
```

```
/* Class selector */
.info {
  color: green;
}
```

```
/* Nested selector */
.container p {
  color: purple;
}
```

G. What is box model in CSS?

The **CSS Box Model** describes how every HTML element is represented as a rectangular box. It defines the space an element occupies and consists of four layers:

1. **Content** – Actual text or image inside the element.
2. **Padding** – Space between content and border; adds inner spacing.
3. **Border** – A line surrounding the padding and content.
4. **Margin** – Space outside the border; creates spacing between elements.

It helps in designing layout, alignment, and spacing of elements accurately.

10. Explain the need of border in designing phase.

The **border** is important in the design phase because:

- It **visually separates** one element from another.
 - Helps in **identifying sections**, boxes, and containers.
 - Useful for creating **focus areas** (ex: forms, cards, buttons).
 - Helps developers **debug spacing** by showing an element's actual size.
 - Enhances **aesthetic appearance**, giving structure to layouts.
 - Used to create effects like outlines, separators, and UI components.
-

11. Write any 2 absolute and any 2 relative measures for CSS properties.

Absolute length units:

1. px → pixels
2. cm → centimeters

(Other examples: mm, in, pt)

Relative length units:

1. em → relative to parent font-size
2. rem → relative to root (HTML) font-size

(Other examples: %, vw, vh)

12. Explain about margin and padding in CSS.

- **Margin:**
 - Space **outside** the border.
 - Creates distance between two elements.
 - Does not affect the element's background color.
 - Example: margin: 20px;
- **Padding:**
 - Space **inside** the border (around content).
 - Increases the clickable/content area.
 - Affects background color because it is inside the box.

- Example: padding: 10px;

Both are essential for layout spacing and readability.

13. Write the HTML document structure.

```
<!DOCTYPE html>
<html>
<head>
  <title>Document Title</title>
</head>
<body>
  <!-- Page Content -->
</body>
</html>
```

14. What is the purpose of HTTPS? How does it differ from HTTP?

Purpose of HTTPS:

HTTPS (HyperText Transfer Protocol Secure) is used to **secure communication** between browser and server. It protects data using **encryption**.

Difference:

HTTP	HTTPS
Not secure	Secure, encrypted
Data can be intercepted	Data is protected using SSL/TLS
Uses port 80	Uses port 443
No certificate	Requires SSL certificate
HTTPS ensures privacy, integrity, and authenticity.	

15. What are web protocols? Name any two and explain briefly.

Web protocols are rules that define how data is transmitted over the internet.

Two important protocols:

1. **HTTP/HTTPS** – Used for transferring webpages between client and server. HTTPS adds encryption.
2. **FTP** – Used for uploading and downloading files between client and server.

Other examples: SMTP, IMAP, TCP, UDP.

16. Explain the client–server architecture used in the web.

Client–server architecture is a model where:

- The **client** (browser) sends a request.
- The **server** processes the request, retrieves data, and sends a response.

Steps:

1. User enters a URL.
2. Browser sends HTTP/HTTPS request to server.
3. Server processes the request.
4. Server sends back HTML/CSS/JS files.
5. Browser displays the webpage.

It separates UI (client) from data processing (server).

17. Design an HTML page for your personal portfolio and apply CSS. Utilize list, table, paragraph tags and apply required layout, color, typography and foundational spacing.

Answer:

```
<!DOCTYPE html>
<html>
<head>
<style>
*{ box-sizing: border-box; }
body{
  font-family:"Segoe UI", Arial, sans-serif;
  color:hsl(210,20%,20%);
  background:hsl(210,40%,97%);
  line-height:1.6;
  margin:20px;
}
h1{
  font-size:2rem;
  font-weight:700;
```

```
    color:rgb(10,90,160);
    border-bottom:3px solid rgba(10,90,160,0.5);
    padding-bottom:5px;
}
```

```
.section{
    margin-bottom:15px;
    padding:15px;
    border:2px solid hsl(210,20%,80%);
    background:white;
    border-radius:8px;
}
```

```
ul{ padding-left:20px; }
```

```
table{
    width:100%;
    border-collapse:collapse;
}
```

```
th,td{
    border:1px solid rgb(180,180,180);
    padding:10px;
}
```

```
th{
    background:hsl(210,30%,90%);
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h1>My Portfolio</h1>
```

```
<div class="section">
```

```
  <h2>About Me</h2>
```

```
<p>Hello! I am a web developer passionate about UI design and coding.</p>
</div>
```

```
<div class="section">
  <h2>Skills</h2>
  <ul>
    <li>HTML, CSS, JavaScript</li>
    <li>Spring Boot C MySQL</li>
    <li>AWS Cloud</li>
  </ul>
</div>
```

```
<div class="section">
  <h2>Projects</h2>
  <table>
    <tr><th>Project</th><th>Year</th></tr>
    <tr><td>Portfolio Website</td><td>2024</td></tr>
    <tr><td>Student Portal</td><td>2023</td></tr>
  </table>
</div>
```

```
</body>
</html>
```

18. Design an HTML page for listing students for the events and apply CSS. Utilize images, table, anchor tags and apply required layout, color, hover and foundational spacing.

Answer:

```
<!DOCTYPE html>
<html>
<head>
<style>
```

```
*{ box-sizing: border-box; }

body{
  font-family:"Calibri",sans-serif;
  background:hsl(100,40%,96%);
  margin:20px;
  color:hsl(120,20%,20%);
  line-height:1.5;
}

h1{
  color:rgb(0,120,60);
  font-size:2rem;
  border-bottom:3px solid rgba(0,120,60,0.5);
  padding-bottom:5px;
}

table{
  width:100%;
  border-collapse:collapse;
  margin-top:15px;
}

th, td{
  border:1px solid rgb(160,160,160);
  padding:10px;
}

th{
  background:hsl(120,20%,90%);
}

a{
  color:rgb(0,100,200);
}

a:hover{
  color:hsl(240,80%,40%);
}
```

```

    text-decoration:underline;
}
img{
    width:60px;
    height:60px;
    border-radius:6px;
}
</style>
</head>

<body>

<h1>Event Students</h1>

<table>
<tr><th>Photo</th><th>Name</th><th>Event</th><th>Profile</th></tr>
<tr><td></td><td>Arjun</td><td>Quiz</td><td><a
href="#">View</a></td></tr>
<tr><td></td><td>Meera</td><td>Design</td><td><a
href="#">View</a></td></tr>
</table>

</body>
</html>

```

1G. Design an HTML page for display the employee information with their salary. Utilize images, paragraphs and apply required layout, color with foundational spacing. Include external CSS to the HTML page.

Answer (HTML + External CSS):

HTML:

```

<!DOCTYPE html>

<html>

```

```
<head>
<link rel="stylesheet" href="style.css">
</head>
<body>

<h1>Employee Details</h1>

<div class="emp-card">
  
  <p><b>Name:</b> Rahul<br><b>Role:</b> Manager<br><b>Salary:</b> ₹75,000</p>
</div>

</body>
</html>
```

style.css:

```
*{ box-sizing:border-box; }

body{
  font-family:Arial,sans-serif;
  background:hsl(200,40%,95%);
  margin:20px;
  color:hsl(200,20%,20%);
  line-height:1.6;
}

h1{
  font-size:2rem;
  color:rgb(0,80,140);
  border-bottom:3px solid rgba(0,80,140,0.4);
  padding-bottom:5px;
}
```

```
.emp-card{
  display:flex;
  gap:20px;
  background:white;
  padding:20px;
  border-radius:8px;
  border:2px solid hsl(200,30%,80%);
}
img{
  border-radius:8px;
}
```

20. Design an HTML page and apply required CSS to show a page with the list of course information for the students. Utilize list, table, div, span and apply required layout, color, typography and foundational spacing. Utilize internal CSS to apply in HTML page.

Answer:

```
<!DOCTYPE html>
<html>
<head>
<style>
*{ box-sizing:border-box; }
body{
  font-family:"Verdana",sans-serif;
  color:hsl(10,20%,20%);
  background:hsl(10,80%,96%);
  margin:20px;
  line-height:1.6;
}
h1{
  color:rgb(200,50,50);
  font-size:2rem;
```

```
border-bottom:3px solid rgba(200,50,50,0.4);
padding-bottom:5px;
}
```

```
.course-box{
border:2px solid hsl(10,30%,80%);
padding:15px;
margin-bottom:15px;
background:white;
border-radius:8px;
}
```

```
span{
color:hsl(10,40%,40%);
font-weight:600;
}
```

```
ul{ padding-left:20px; }
```

```
table{
width:100%;
border-collapse:collapse;
}
```

```
th,td{
border:1px solid rgb(180,180,180);
padding:10px;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h1>Courses</h1>
```

```
<div class="course-box">
```

```
<p><span>Course:</span> Web Development</p>
```

```
<ul>
```

```
<li>HTML</li>
```

```
<li>CSS</li>
```

```
<li>JS</li>
```

```
</ul>
```

```
<table>
```

```
<tr><th>Module</th><th>Hours</th></tr>
```

```
<tr><td>Frontend Basics</td><td>40</td></tr>
```

```
<tr><td>Backend Basics</td><td>50</td></tr>
```

```
</table>
```

```
</div>
```

```
</body>
```

```
</html>
```

21. Create a homepage for a college website with headings, navigation links, and styled content using basic CSS. Apply external CSS to HTML page.

Answer (HTML + CSS):

HTML:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<link rel="stylesheet" href="college.css">
```

```
</head>
```

```
<body>
```

```
<header>
```

```
<h1>ABC College</h1>
```

```
<nav>
```

```
<a href="#">Home</a>
<a href="#">Departments</a>
<a href="#">Admissions</a>
<a href="#">Contact</a>
</nav>
</header>
```

```
<section class="info">
<h2>Welcome</h2>
<p>Leading institution with quality education for students.</p>
</section>
```

```
</body>
</html>
```

college.css:

```
*{ box-sizing:border-box; }
```

```
body{
    font-family:Arial,sans-serif;
    background:hsl(230,40%,96%);
    margin:20px;
    color:hsl(230,20%,20%);
}
```

```
header{
    border-bottom:3px solid rgba(0,0,0,0.2);
    padding-bottom:10px;
}
```

```
nav a{
    margin-right:20px;
    color:rgb(50,80,200);
}
```

```
text-decoration:none;
}
nav a:hover{
  color:hsl(240,70%,40%);
}
```

22. Build an article layout using headings, subheadings, paragraphs, and style it using fonts, line spacing, and margin/padding. Apply internal CSS to HTML page.

Answer:

```
<!DOCTYPE html>
<html>
<head>
<style>
*{ box-sizing:border-box; }
body{
  font-family:"Georgia",serif;
  background:hsl(50,40%,95%);
  margin:30px;
  color:hsl(50,20%,20%);
  line-height:1.8;
}
.container{
  padding:25px;
  border:2px solid rgba(0,0,0,0.2);
  border-radius:10px;
  background:white;
}
h1{
  font-size:2.4rem;
  color:rgb(150,100,0);
  letter-spacing:1px;
```

```

    margin-bottom:15px;
}
h2{
    font-size:1.8rem;
    color:hsl(50,30%,30%);
    margin-top:20px;
    margin-bottom:10px;
}
p{
    margin-bottom:15px;
    text-align:justify;
}
</style>
</head>

<body>
<div class="container">
<h1>Impact of Technology</h1>
<h2>Introduction</h2>
<p>Technology affects all aspects of life, improving communication, healthcare, and education globally.</p>
<h2>Applications</h2>
<p>Automation, AI, and cloud computing are transforming industries and daily life.</p>
</div>
</body>
</html>

```

23. Build an HTML page that uses nested content (headings, lists, links) to display student info and style it using padding, margin, and border to create structured spacing.

Answer:

```

<!DOCTYPE html>

<html>

```

```
<head>
<style>
*{ box-sizing:border-box; }
body{
  font-family:"Arial",sans-serif;
  background:hsl(200,40%,96%);
  margin:20px;
  color:hsl(200,20%,20%);
}
.container{
  border:2px solid rgba(0,0,0,0.2);
  padding:20px;
  border-radius:8px;
  background:white;
}
ul{
  padding-left:25px;
  margin-top:10px;
}
li{
  margin-bottom:8px;
}

a{
  color:rgb(0,80,200);
  text-decoration:none;
}
a:hover{
  color:hsl(210,80%,40%);
  text-decoration:underline;
}
</style>
```

```
</head>

<body>

<div class="container">
<h2>Student Profile</h2>
<ul>
<li>Name: Naveen</li>
<li>Department: CSE</li>
<li>Email: <a href="#">naveen@example.com</a></li>
<li>Interests:
  <ul>
    <li>AI</li>
    <li>Web Development</li>
  </ul>
</li>
</ul>
</div>

</body>
</html>
```

24. Build a basic webpage that displays a product description using headings, paragraphs, links, and simple CSS color/typography and foundational space styling.

Answer:

```
<!DOCTYPE html>

<html>
<head>
<style>
*{ box-sizing:border-box; }
body{
```

```
font-family:"Segoe UI",sans-serif;
background:hsl(340,40%,96%);
margin:20px;
color:hsl(340,20%,20%);
line-height:1.6;
}
```

```
h1{
font-size:2rem;
font-weight:700;
color:rgb(180,0,90);
margin-bottom:10px;
}
```

```
p{
margin-bottom:15px;
}
```

```
a{
color:hsl(340,80%,40%);
text-decoration:none;
}
```

```
a:hover{
color:rgb(220,0,120);
text-decoration:underline;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h1>Smartwatch X</h1>
```

```
<p>Advanced smartwatch with heart-rate monitoring, GPS, and water resistance.</p>
```

```
<p><strong>Price:</strong> ₹4999</p>
```

<p>More details: Click Here</p>

</body>

</html>

25. What is the use of the <textarea> tag?

Answer:

It is used to accept multi-line text input from the user. Mention any two media tags in HTML.

26. What is the purpose of the alt attribute?

Answer:

It provides alternative text for images, useful for accessibility and screen readers.

27. What does the CSS: hover selector do?

Answer:

It applies styles when the user places the mouse pointer over an element.

28. Explain different form elements used in HTML.

Answer:

HTML forms use elements like <input> for single-line input, <textarea> for multi-line text, <select> and <option> for dropdown lists, and <button> to submit or reset forms.

29. Describe any four HTML input types.

Answer:

text: Accepts plain text

email: Accepts valid email addresses

number: Accepts numeric values

password: Hides user input

30. Write HTML code to validate a 12-digit Aadhaar number.

<html>

<head>

<title>Aadhaar Validation</title>

</head>

<body>

<h2>Aadhaar Number Validation</h2>

<form>

```
<label>Aadhaar Number:</label><br>
<input type="text" placeholder="Enter 12-digit Aadhaar" pattern="[0-9]{12}"
  required><br><br>

  <button type="submit">Verify</button>
</form>

</body>
</html>
```

31. Write the pattern for Indian mobile number validation.

```
<!DOCTYPE html>
<html>
<head>
  <title>Mobile Number Validation</title>
</head>
<body>

  <h2>Mobile Number Validation</h2>

  <form>
    <label>Mobile Number:</label><br>
    <input type="text" placeholder="Enter mobile number" pattern="[0-9]{10}"
      required><br><br>

    <button>Submit</button>
  </form>

</body>
</html>
```